

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 5: Feedback Loops: How the Body Keeps Water and Blood Glucose Steady

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	C	C	D	B	C

USE AFTER: PREDICT PHASE — Predict: Wanjiku scenario (Think-Pair-Share) prediction about urine volume and concentration

Q1. Wanjiku runs 10 km on a hot afternoon in Kericho. She sweats a lot and drinks very little water, so her blood becomes more concentrated. What will her kidneys most likely produce?

- A. A small volume of concentrated urine, to save body water**
- B. Urine with a lot of glucose, because running uses sugar
- C. A large volume of dilute urine, to remove the extra water
- D. The same amount of urine as normal, because the kidney does not respond to the body's water level

Correct answer: A. Concentrated blood means the body is short of water, so more ADH is released and more water is reabsorbed. This gives a small volume of concentrated urine.

USE AFTER: OBSERVE PHASE — Part A: osmoregulation flowchart (ADH, collecting duct) built from the homeostasis feedback charts

Q2. In osmoregulation, more ADH is released when the blood is too concentrated. Why does more ADH lead to less urine?

- A. ADH makes the bladder store water so it is not passed out
- B. ADH makes the glomerulus filter less blood, so no water enters the nephron
- C. ADH is a waste product that carries water out of the body
- D. ADH makes the collecting ducts more permeable, so more water is reabsorbed into the blood**

Correct answer: D. ADH acts on the collecting duct and makes it more permeable to water. More water returns to the blood, leaving a small volume of concentrated urine.

USE AFTER: OBSERVE PHASE — Part B: blood glucose flowchart (high glucose after a meal → beta cells → insulin)

Q3. After eating a big plate of ugali, a person's blood glucose rises above its set point. Which statement correctly describes the negative feedback response?

- A. The kidney alone decides to filter out the extra glucose, and no hormone is needed
- B. Insulin removes sugar from the body by carrying it out in the urine
- C. Beta cells in the pancreas release insulin, so liver and muscle take up glucose and store it as glycogen, and blood glucose falls back**
- D. Alpha cells release glucagon, which makes the liver release more glucose into the blood

Correct answer: C. Negative feedback detects a change from the set point and reverses it. High glucose triggers insulin from beta cells. Insulin moves glucose into cells and stores it as glycogen, so the level returns to normal.

USE AFTER: OBSERVE PHASE — Part B: blood glucose flowchart (low glucose after a long run or fasting → alpha cells → glucagon → liver)

Q4. A runner's blood glucose falls low after a long run. Which hormone is released, and which organ responds to bring blood glucose back up?

- A. Glucagon from the beta cells; the muscles change glucose into urine
- B. ADH from the pituitary; the collecting duct releases glucose into the blood
- C. Glucagon from the alpha cells; the liver changes glycogen into glucose**
- D. Insulin from the alpha cells; the kidney changes glucose into glycogen

Correct answer: C. Low blood glucose makes alpha cells in the islets of Langerhans release glucagon. Glucagon makes the liver convert stored glycogen into glucose, which raises blood glucose.

USE AFTER: OBSERVE PHASE — Part C: case study 2, diabetes insipidus (patient in Kisumu), marking where the loop is broken

Q5. A patient in Kisumu passes up to 10 litres of very dilute urine each day and is always thirsty. Tests show normal blood glucose and no glucose in the urine. Which step in the body's control system has most likely failed?

- A. Beta cells release too little insulin
- B. The proximal convoluted tubule cannot reabsorb glucose
- C. Alpha cells release too much glucagon
- D. ADH is missing, or the kidney does not respond to it, so too little water is reabsorbed**

Correct answer: D. Large volumes of dilute urine with normal blood glucose point to diabetes insipidus. ADH is low or ineffective, so the collecting ducts do not reabsorb enough water.

USE AFTER: EXPLAIN PHASE — Explain: annotating the nephron diagram to show where glucose is reabsorbed (proximal convoluted tubule) and why glucose appears in urine in diabetes mellitus

Q6. Normally all the glucose that is filtered into a nephron is reabsorbed in the proximal convoluted tubule, so urine has no glucose. Why does glucose appear in the urine of a person with diabetes mellitus?

- A. The kidney makes glucose from water when ADH is low
- B. Too little insulin makes blood glucose very high, so more glucose is filtered than the tubule can reabsorb**
- C. Too much ADH stops the collecting duct from reabsorbing glucose
- D. Glucagon destroys the glucose before it reaches the nephron

Correct answer: B. In diabetes mellitus, insulin is missing or ineffective, so blood glucose stays high. The proximal convoluted tubule cannot reabsorb all the filtered glucose, so some is passed out in the urine.

USE AFTER: EXPLAIN PHASE — Explain: applying the reasoning to the goat and the freshwater fish (Predict-Observe-Explain written explanation)

Q7. A goat on a hot afternoon produces concentrated urine, but a freshwater fish produces large amounts of dilute urine. Which explanation is best?

- A. Both animals produce the same urine, but the fish looks like it makes more because it lives in water
- B. The goat has a high blood sugar level, which makes its urine concentrated
- C. The goat conserves water, with high ADH and more water reabsorbed, while the fish gains water by osmosis and must excrete the extra as dilute urine**
- D. The fish has no kidneys, so its urine is just water from its gills

Correct answer: C. The goat conserves water, so ADH is high and its urine is concentrated. Water enters the freshwater fish by osmosis, so it excretes a large volume of dilute urine.